

BIOCODE ASSIGNMENT 02

NAMRA MAJEED

Question 1: Retrieving Complete Information from a Gene Accession Number

Step-by-Step Approach Using NCBI Resources

When provided with a gene accession number, the most efficient and comprehensive retrieval strategy leverages NCBI's Entrez system, which integrates over 40 molecular biology databases through a unified search interface.

1. Initial Entry Point, Nucleotide Database

Begin by navigating to the NCBI Nucleotide database and entering the accession number. This returns the complete DNA sequence record, including coding sequence (CDS) annotations, exon-intron boundaries, and flanking regions. The record displays the gene's genomic context, providing critical structural information about the locus.

2. Protein Sequence Retrieval

From the nucleotide record, I would either:

- Click on the "Protein" link in the "Related Information" section to access the translated protein sequence in the Protein database
- Alternatively, use the accession number with the "protein" qualifier in the search bar.

The Protein database entry provides the amino acid sequence, conserved domain annotations, and computational predictions about protein function. This is essential for understanding the gene product's potential biological role.

3. Associated Literature via PubMed

Within the nucleotide record, I would select the "PubMed" link to retrieve all scientific publications referencing this specific sequence. The Gene database record also provides a curated list of publications relevant to the gene's function, expression, and clinical significance. This is critical for understanding the biological context, experimental evidence, and known variants.

4. Gene Information via Gene Database

Directly searching the Gene database with the accession number provides comprehensive annotations: official gene symbol and name, chromosomal location, Gene Ontology (GO) terms for function and cellular localization, phenotype descriptions, and links to disease associations. The Gene record integrates and distills information from multiple sources into a concise summary.

Justification for Database Selection

Each NCBI database serves a distinct purpose: Nucleotide provides the primary sequence blueprint, Protein offers the translated functional product, PubMed delivers the experimental and

theoretical knowledge foundation, and Gene provides the curated biological context and nomenclature. Using these resources sequentially ensures complete information retrieval from raw sequence data to functional understanding.

Question 2: Studying the 3D Structure of Human Insulin

Recommended Database: Protein Data Bank (PDB)

The Protein Data Bank is the definitive global repository for experimentally determined three-dimensional (3D) structures of biological macromolecules, including proteins and nucleic acids. For studying human insulin structure-function relationships, PDB is uniquely suited because it provides atomic-resolution structural data derived from X-ray crystallography, NMR spectroscopy, and cryo-electron microscopy.

Retrieval Strategy

To retrieve the human insulin structure:

1. Navigate to the PDB website (rcsb.org) and use the search function
2. Enter "human insulin" as the query, or use the UniProt accession number **P01308** to find all experimentally determined structures of insulin
3. Alternatively, search by specific PDB identifiers such as **1HIT** (a well-characterized mutant insulin structure) or **1QJ0** (insulin hexamer)

Information Available in PDB to Support Research

The PDB entry provides multiple data layers essential for structure-function studies:

1. **Atomic Coordinates:** Precise 3D positions of every atom in the protein, enabling visualization of the insulin structure and its two-chain architecture (A-chain and B-chain) connected by disulfide bridges. This permits detailed structural analysis of how mutations may disrupt conformation.
2. **Experiment Details:** Information about the method used (X-ray resolution, NMR conditions) provides context for data quality and reliability. For example, the validation report includes metrics such as clash scores and Ramachandran outliers that indicate structural quality.
3. **Cross-references to UniProt:** The PDB entry links directly to UniProt P01308, enabling integration of structural data with sequence, function, and disease annotations. This is vital for understanding how specific mutations listed in UniProt map onto the 3D structure.

4. **Ligand and Cofactor Information:** For insulin, PDB entries show zinc ions and water molecules that are biologically relevant for hormone storage and stability.
5. **Structural Neighbors and Classification:** PDB provides structural classification (e.g., hormone/growth factor) and lists of similar structures, enabling comparative analysis across species or mutants to understand how specific amino acid substitutions affect conformation.

By analyzing how mutations alter insulin's structure in PDB, researchers can predict whether a given mutation compromises receptor binding, stability, or proper folding—directly informing understanding of functional consequences.

Question 3: Complete Workflow for Analyzing a Protein from Accession ID Only

Comprehensive Workflow Using NCBI, UniProt, and PDB

When beginning with only a protein accession ID, a systematic multi-database approach ensures comprehensive characterization of the protein's identity, function, structure, and biological significance.

Step 1: Primary Identification and Functional Annotation, UniProt

Start with UniProtKB by entering the accession ID. UniProt acts as the central hub of expert-curated protein information, providing the most comprehensive functional annotation.

- Retrieve the protein's **function** from the "Function" section, which includes catalytic activity, binding partners, and biological processes based on experimental evidence
- Access the **Gene name** and **organism** to establish biological context
- Review **domain architecture** and **post-translational modifications** that regulate activity
- Identify **disease associations** and natural variants that may impact function
- Examine **subcellular localization** to understand where the protein acts

Step 2: Sequence and Genomic Context, NCBI

Move to NCBI's Protein database using the same accession ID:

- Access the **complete amino acid sequence** in FASTA format for downstream analysis
- Retrieve **conserved domain annotations** through CD-search, which identifies functional modules and evolutionary relationships
- Link to the corresponding **nucleotide record** to examine gene structure, regulatory elements, and genomic context

- Access the **Gene database** for official nomenclature, chromosomal location, and phenotype information

Step 3: Structural Information, Protein Data Bank

Query PDB using the UniProt accession ID (cross-referenced through SIFTS mapping):

- Retrieve **experimentally determined 3D structures**, if available
- Examine **atomic coordinates** to visualize the protein's conformation
- Identify **binding sites**, **active sites**, and **ligand interactions** that define function
- Access **validation reports** to assess structure quality

If no experimental structure exists, use the UniProt accession to search for **homology models** or use the sequence for **structure prediction** via AlphaFold or related resources.

Step 4: Integration and Synthesis

Each database provides complementary information:

- **UniProt** offers functional annotation and biological knowledge (the "what and why")
- **NCBI** provides sequence data and genomic context (the "where and how")
- **PDB** provides 3D architecture explaining how structure supports function (the "physical basis")

This integrated workflow transforms a simple accession ID into a complete biological narrative, revealing the protein's role in cellular processes, its structural basis, and its potential clinical significance.

PART 2: SEQUENCE RETRIEVAL EXERCISE

Retrieved Database Record

Accession ID: P01308

Entry Name: INS_HUMAN

Gene Name: INS

Protein Name: Insulin

Source Organism: *Homo sapiens* (Human)

Screenshot of Database Record

The screenshot shows the UniProtKB entry for P01308 (INS_HUMAN). The page layout includes a left sidebar with navigation links (Function, Names & Taxonomy, Subcellular Location, Disease & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence & Isoform) and a main content area. The main content area displays the following information:

Category	Value
Protein	Insulin
Gene	INS
Status	UniProtKB reviewed (Swiss-Prot)
Organism	Homo sapiens (Human)
Amino acids	110 (go to sequence)
Protein existence	Evidence at protein level
Annotation score	5/5

Below the main content area, there are tabs for Entry, Variant viewer (180), Feature viewer, Genomic coordinates, Publications, External links, and History. The 'Function' tab is currently selected, showing the text 'Function:'. At the bottom of the page, there is a cookie consent banner and a Windows taskbar with various application icons and system clock showing 10:27 PM on 7/17/2025.

- **Accession number:** P01308
- **Entry name:** INS_HUMAN
- **Protein name:** Insulin
- **Gene:** INS
- **Organism:** Homo sapiens
- **Length:** 110 amino acids
- **Function:** Insulin decreases blood glucose concentration...

Biological Description of Human Insulin

What is Human Insulin?

Human insulin (UniProt P01308) is a peptide hormone produced by the beta cells of the pancreatic islets of Langerhans. It is synthesized as a single-chain precursor, preproinsulin, which undergoes proteolytic processing to yield the active hormone consisting of two polypeptide chains: the A-chain (21 amino acids) and B-chain (30 amino acids). These chains are

linked by two interchain disulfide bonds, with an additional intrachain disulfide bridge in the A-chain, stabilizing the three-dimensional structure essential for biological activity.

Biological Function

Insulin is the primary anabolic hormone responsible for regulating glucose homeostasis. It functions by binding to the insulin receptor (IR) on target cells, triggering a signaling cascade that promotes glucose uptake from the bloodstream into skeletal muscle and adipose tissue. Simultaneously, insulin stimulates glycogenesis, lipogenesis, and protein synthesis while inhibiting gluconeogenesis and lipolysis. Through this dual mechanism of promoting storage and preventing breakdown of energy reserves, insulin maintains blood glucose concentrations within a narrow physiological range essential for normal cellular function, particularly for glucose-dependent tissues such as the brain.

Why It Is Important

Insulin is medically significant because its deficiency or dysfunction leads to diabetes mellitus, a global epidemic affecting over 500 million people worldwide. Type 1 diabetes results from autoimmune destruction of beta cells causing absolute insulin deficiency, while Type 2 diabetes involves insulin resistance combined with relative insulin deficiency. Understanding insulin's structure-function relationships has enabled development of recombinant insulin therapies, insulin analogs with modified pharmacokinetics, and novel drug delivery systems. Over 100 insulin mutations have been characterized, many associated with monogenic diabetes (MODY) or hyperinsulinemic hypoglycemia, making it a critical model system for studying protein misfolding diseases and structure-guided drug design.

Database Selection

I selected **UniProt (accession P01308)** as the primary database for this retrieval because it provides the most comprehensive and expertly curated protein-specific information available. UniProt integrates functional annotations, post-translational modifications, disease associations, and cross-references to structural databases (PDB) and genomic resources (NCBI). The entry presents information in an accessible yet scientifically rigorous format, with evidence tags that distinguish experimentally validated data from computational predictions. This holistic view of insulin's molecular properties, biological role, and clinical significance demonstrates why UniProt serves as the ideal starting point for protein characterization.